A Deep Dive into Fokker-Planck

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Reference (1)

Intro

We start this note by stating out the Fokker–Planck Equation.

Fokker-Planck Equation

Suppose the dynamics of X_t can be described by a forward SDE

$$dX_t = f(X_t, t)\,dt + g(t)\,dW_t.$$

If the marginal of X_t (the distribution of X at time t) is known $p_t(X_t)$ then the Fokker–Planck equation gives the PDE of the marginal:

$$\partial_t p_t(x) = -\nabla \cdot \big(f(x,t)\,p_t(x)\big) + \tfrac{1}{2} g^2(t)\,\Delta p_t(x).$$

People everywhere just cited this as if it's a basic facts any educated ones should know. Let us become educated today.

Core Concept: Change of Variable

If we recall our freshman year Probability, we're already familiar with the density distribution after a diterministic mapping of a random variable.

Derived Distribution (Discrete Transition)

Suppose a continuous random variable $\mathbf{x}_0 \sim p_0$ was mapped by a deterministic state transition function $\boldsymbol{\Psi} : \mathbb{R}^n \to \mathbb{R}^n$.

$$\mathbf{x}_1 = \boldsymbol{\Psi}(\mathbf{x}_0).$$

Then the pdf of $\mathbf{x}_1$ is

$$p_1(\mathbf{x}_1) = p_0(\boldsymbol{\Psi}^{-1}(\mathbf{x}_1)) \left| \det \left(\frac{\partial \boldsymbol{\Psi}^{-1}}{\partial \mathbf{x}_1} \right) \right|.$$

We suppose everything is nice in here (Song is a physicist.)

Continuous State Transition (1)

Let us now consider a continuous time case. Now the state transition is described by an ODE,

$$d\mathbf{x}_t = \mathbf{f}(\mathbf{x}_t, t)\, dt.$$

where the state transition is now described by the function $\mathbf{f} : \mathbb{R}^n \times [0, T] \to \mathbb{R}^n$.

Derived Distribution (Continuous Transition)

For a state transition following an ODE $d\mathbf{x}_t = \mathbf{f}(\mathbf{x}_t, t)\, dt$, the probability distribution of $\mathbf{x}$ evolves as

$$\frac{\partial}{\partial t} p_t(\mathbf{x}) = -\nabla \cdot (p_t(\mathbf{x}) \mathbf{f}(\mathbf{x}, t)).$$

Continuous State Transition (2)

The proof starts from equivalently interpret the state transition ODE as small bijective updates

$$\mathbf{x}_{t+\Delta t} = \boldsymbol{\Psi}(\mathbf{x}_t) \coloneqq \mathbf{x}_t + \Delta t \mathbf{f}(\mathbf{x}_t, t). \tag{1}$$

Treating it as nice deterministic function as in the previous section, we find its Jacobian,

$$\begin{aligned}
\frac{\partial \Psi}{\partial \mathbf{x}_t} &= I + \Delta t \, \nabla_{\mathbf{x}} f(\mathbf{x}_t, t) + O(\Delta t^2), \\
\det\left(\frac{\partial \Psi}{\partial \mathbf{x}_t}\right) &= \det(I + \Delta t \, \nabla_{\mathbf{x}} f(\mathbf{x}_t, t)) + O(\Delta t^2) \\
&= 1 + \Delta t \, \mathrm{Tr}\left(\nabla_{\mathbf{x}} f(\mathbf{x}_t, t)\right) + O(\Delta t^2) \\
&= 1 + \Delta t \, \nabla_{\mathbf{x}} \cdot f(\mathbf{x}_t, t) + O(\Delta t^2).
\end{aligned}$$

Continuous State Transition (3)

And thus the log density follows, if we look at only the first order terms w.r.t. Δt

$$\begin{aligned}\log p_{t+\Delta t}(x_{t+\Delta t}) &= \log p_t(\mathbf{x}_t) - \log(1 + \Delta t\, \nabla_{\mathbf{x}} \cdot f(\mathbf{x}_t, t)) + O(\Delta t^2) \\ &= \log p_t(\mathbf{x}_t) - \Delta t\, \nabla_{\mathbf{x}} \cdot f(\mathbf{x}_t, t) + O(\Delta t^2).\end{aligned}$$

And therefore,

$$\log p_{t+\Delta t}(x_{t+\Delta t}) - \log p_t(x_t) = -\Delta t\, \nabla_x \cdot f(x_t, t) + O(\Delta t^2). \quad \text{(B.1.5)}$$

Let us now starts from the first order Taylor expansion of the LHS,

$$\begin{aligned}&\log p_{t+\Delta t}(\mathbf{x}_{t+\Delta t}) - \log p_t(\mathbf{x}_t) \\ &\quad = \Delta t\, \partial_t \log p_t(x_t) + (\mathbf{x}_{t+\Delta t} - \mathbf{x}_t)^\top \nabla_x \log p_t(\mathbf{x}_t) + O(\Delta t^2) \\ &\quad = \Delta t\, \partial_t \log p_t(\mathbf{x}_t) + \Delta t\, f(\mathbf{x}_t, t)^\top \nabla_{\mathbf{x}} \log p_t(\mathbf{x}_t) + O(\Delta t^2).\end{aligned}$$

Continuous State Transition (4)

Matching this with the RHS of (B.1.5) and letting $\Delta t \to 0$ yields

$$\partial_t \log p_t(\mathbf{x}_t) = -\nabla_{\mathbf{x}} \cdot f(\mathbf{x}_t, t) - f(\mathbf{x}_t, t)^\top \nabla_{\mathbf{x}} \log p_t(\mathbf{x}_t). \tag{2}$$

lifting log by chain rule

$$\begin{aligned}
\partial_t p_t(\mathbf{x}_t) &= p_t(\mathbf{x}_t) \partial_t \log p_t(\mathbf{x}_t) \\
&= p_t(\mathbf{x}_t)[-\nabla_{\mathbf{x}} \cdot f(\mathbf{x}_t, t) - f(\mathbf{x}_t, t)^\top \nabla_{\mathbf{x}} \log p_t(\mathbf{x}_t)] \\
&= [-p_t(\mathbf{x}_t) \nabla_{\mathbf{x}} \cdot f(\mathbf{x}_t, t) - f(\mathbf{x}_t, t)^\top p_t(\mathbf{x}_t) \nabla_{\mathbf{x}} \log p_t(\mathbf{x}_t)] \\
&= [-p_t(\mathbf{x}_t) \nabla_{\mathbf{x}} \cdot f(\mathbf{x}_t, t) - f(\mathbf{x}_t, t)^\top \nabla_{\mathbf{x}} \cdot p_t(\mathbf{x}_t)] \\
&= -\nabla_{\mathbf{x}} \cdot (p_t(\mathbf{x}_t) f(\mathbf{x}_t, t))
\end{aligned}$$

Density Evolution of a Continuous State Transition (1)

Continuity Equation

Let $\mathbf{f} : \mathbb{R}^n \times [0, T] \to \mathbb{R}^n$ be a time-dependent mapping satisfying

- for each time slices $f(\cdot, t) \in C^1$
- f is locally Lipschitz in x

Consider the states $\mathbf{x}_t$,$t \in [0, T]$ defined by the deterministic state transition,

$$d\mathbf{x}_t = \mathbf{f}(\mathbf{x}_t, t)\, dt,$$

with a given initial state $x_0 \sim p_0$.

Assume that for each time slice t, the law of the $\mathbf{x}_t$ follows a nice density p_t. Then the family of the densities of all t satisfies the Continuity Equation

$$\frac{\partial}{\partial t} p_t(\mathbf{x}) = -\nabla_{\mathbf{x}} \cdot (p_t(\mathbf{x}) \mathbf{f}(\mathbf{x}, t)).$$

Density Evolution of a Continuous State Transition (2)

Let us try to inspect its physical intuition.

- the overall "probability mass" is conservative.
- the drift function (the velocity field) acts on the particle $\mathbf{x}_t$ only by re-distributing them in space.
- the divergence $\nabla_{\mathbf{x}} \cdot (p_t(\mathbf{x})\mathbf{f}(\mathbf{x}, t))$ described how the flow stretches or compresses the density
- correspondingly the density changes to preserve the total probability.

Deterministic vs. Stochastic Dynamics and Density Evolution

Let us put them together and try to gain some intuition.

Deterministic	**Stochastic**
$d\mathbf{x}_t = \mathbf{f}(\mathbf{x}_t, t)dt$	$d\mathbf{x}_t = \mathbf{f}(\mathbf{x}_t, t)\,dt + g(t)\,dW_t$
$\partial_t p_t(\mathbf{x}) = -\nabla_{\mathbf{x}} \cdot \big(p_t(\mathbf{x})\mathbf{f}(\mathbf{x}, t)\big)$	$\partial_t p_t(\mathbf{x}) = -\nabla_{\mathbf{x}} \cdot \big[p_t(\mathbf{x})\big(\mathbf{f}(\mathbf{x}, t) + \frac{1}{2}g^2(t)\nabla_{\mathbf{x}} \log p_t(\mathbf{x})\big)\big]$

The intuition is direct here. We know the Brownian increment $d\mathbf{W}_t \sim \mathcal{N}(0, dt)$ satisfies

$$(d\mathbf{W}_t)^2 = dt$$

and thus in the Taylor expansion we need to consider up to second order in stochastic calculus, therefore the additional diffusion term emerged.

Chain Rules from Taylor Expansion

Let us start from the chain rule for a deterministic process.

Consider x_t following some ODE. For a nice function $h(x_t, t)$ if we want to inspect its evolution w.r.t. time,

$$dh = \left(\frac{\partial h}{\partial t} + \frac{\partial h}{\partial x_t} f\right) dt$$

We write the Taylor expansion up to second order term,

$$\begin{aligned} h(x_{t+dt},\, t+dt) - h(x_t,\, t) = &\frac{\partial h}{\partial t}\, dt + \frac{\partial h}{\partial x}\, dx_t \\ &+ \frac{1}{2}\left(\frac{\partial^2 h}{\partial x^2}(dx_t)^2 + 2\,\frac{\partial^2 h}{\partial t \partial x}\, dt\, dx_t + \frac{\partial^2 h}{\partial t^2}\,(dt)^2\right) \end{aligned}$$

Stochastic Chain Rules (Ito's Lemma) (1)

Let us now consider x_t following an SDE,

$$dx_t = f(x_t, t)\, dt + g(t)\, dw_t$$

We still can have the same Taylor expansion, but we need to inspect the cross terms carefully since they don't easily vanishes w.r.t. dt.

- **Negligible Cross Terms $dt\, dx_t$ and $dt\, dt$.**
  By Brownian motion,

$$dw_t = O(\sqrt{dt}) \quad \Longrightarrow \quad dt \cdot dw_t = O\Big((dt)^{3/2}\Big).$$

$$dt\, dx_t = dt\,(f\, dt + g\, dw_t) = f(dt)^2 + g\, dt \cdot dw_t = O\Big((dt)^{3/2}\Big).$$

$$dt\, dt = O\big((dt)^2\big).$$

  Thus they vanish.

Stochastic Chain Rules (Ito's Lemma) (2)

- **Second-Order Term $(dx_t)^2$.** Expanding using the SDE,

$$\begin{aligned}(dx_t)^2 &= (f\,dt + g\,dw_t)^2 \\ &= f^2(dt)^2 + 2fg\,dt\,dw_t + g^2(dw_t)^2 \\ &= O\big((dt)^2\big) + O\Big((dt)^{3/2}\Big) + g^2\,O(dt) \\ &= g^2(t)\,dt + O\Big((dt)^{3/2}\Big).\end{aligned}$$

- We therefore obtain the differential,

$$dh(x_t,t) = \frac{\partial h}{\partial t}\,dt + \frac{\partial h}{\partial x}\,dx_t + \frac{1}{2}\,\frac{\partial^2 h}{\partial x^2}\,g^2(t)\,dt.$$

Substituting

$$dx_t = f(x_t,t)\,dt + g(t)\,dw_t,$$

yields

Stochastic Chain Rules (Ito's Lemma) (3)

$$dh(x_t, t) = \left(\frac{\partial h}{\partial t} + f \frac{\partial h}{\partial x} + \frac{1}{2} g^2 \frac{\partial^2 h}{\partial x^2} \right) dt \; + \; g \frac{\partial h}{\partial x} \, dw_t.$$

This is the 1D Itô's formula. The multivariate case follows immediately.

Stochastic Chain Rules (Ito's Lemma) (4)

Multidimensional Itô Formula

Let $\mathbf{x}_t \in \mathbb{R}^D$ satisfy the SDE

$$d\mathbf{x}_t = f(\mathbf{x}_t, t)\, dt + g(t)\, d\mathbf{w}_t,$$

where $f : \mathbb{R}^n \times [0, T] \to \mathbb{R}^n$, $g : [0, T] \to \mathbb{R}$, and $\mathbf{w}_t \in \mathbb{R}^n$ is an n-dimensional Brownian motion.
Let $h : \mathbb{R}^n \times [0, T] \to \mathbb{R}$ be a scalar-valued function. Then $h(\mathbf{x}_t, t)$ satisfies:

$$dh(\mathbf{x}_t, t) = \left(\frac{\partial h}{\partial t} + \nabla_{\mathbf{x}} h^\top f + \frac{1}{2} g^2(t) \operatorname{Tr}\left(\nabla_{\mathbf{x}}^2 h\right) \right) dt \; + \; g(t)\, \nabla_{\mathbf{x}} h^\top d\mathbf{w}_t.$$

Deriving the Fokker–Planck Equation (1)

Let $\phi(x,t)$ be a smooth test function $\phi : \mathbb{R}^n \times [0,T] \to \mathbb{R}$.

By Itô's formula, we know the derivative

$$d\phi(x_t,t) = \left(\frac{\partial \phi}{\partial t} + \nabla_x \phi^\top f(x_t,t) + \frac{1}{2} g^2(t) \operatorname{Tr}\left[\nabla_x^2 \phi\right] \right) dt + g(t)\, \nabla_x \phi^\top dw_t.$$

Taking expectation over the law p_t and using $\mathbb{E}[dw_t] = 0$,

$$\begin{aligned} \mathbb{E}[d\phi(x_t,t)] &= \mathbb{E}\left[\left(\frac{\partial \phi}{\partial t} + \nabla_x \phi^\top f(x_t,t) + \frac{1}{2} g^2(t) \operatorname{Tr}(\nabla_x^2 \phi) \right) dt \right]. \\ &= \int \left(\frac{\partial \phi}{\partial t} + \nabla_x \phi^\top f(x,t) + \frac{1}{2} g^2(t) \operatorname{Tr}(\nabla_x^2 \phi) \right) p_t(x)\, dx\, dt. \end{aligned}$$

Deriving the Fokker–Planck Equation (2)

If we dare to recall our Vector Calculus, using divergence theorem,

$$\int \nabla_x \phi^\top f \, p_t \, dx = -\int \phi \, \nabla_x \cdot (f p_t) \, dx,$$
$$\int \mathrm{Tr}(\nabla_x^2 \phi) \, p_t \, dx = -\int \phi \, \Delta p_t \, dx.$$

Plugging back to the expectation and we obtain,

$$\mathbb{E}[d\phi(x_t, t)] = \int \phi(x, t) \left[\frac{\partial p_t}{\partial t} + \nabla_x \cdot (f p_t) - \frac{1}{2} g^2(t) \, \Delta p_t \right] dx \, dt.$$

Since ϕ is arbitrary, the integrand must vanish. Setting the terms inside the parenthesis and we arrived at,

$$\frac{\partial p_t}{\partial t} = -\nabla_x \cdot (f(x, t) \, p_t(x)) + \frac{1}{2} g^2(t) \, \Delta p_t(x).$$